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# Computationally Efficient Multivariate Changepoint Detection with Subsets

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Joint work with Ben Pickering & Sean Ryan
Maastricht, 2020

# Outline



- Motivation
- Current Approaches

- Modelling Subset-Multivariate Changepoints
- Subset Multivariate Optimal Partitioning
- Example

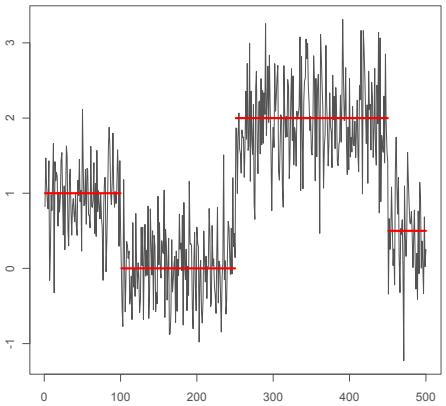
- Approximate but computationally efficient implementation
- Simulations
- Examples

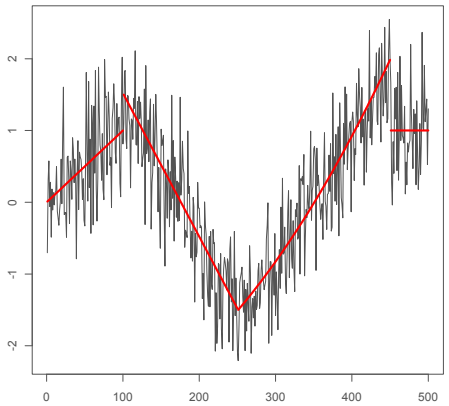
# Scene setting: What is changepoint analysis?

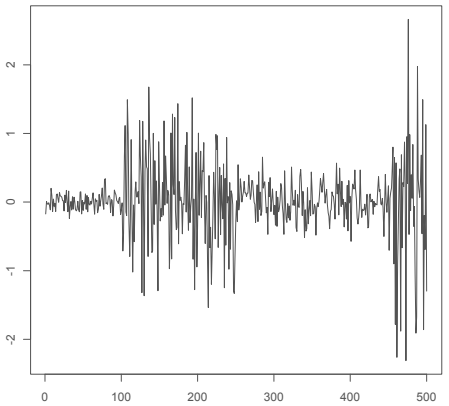
# What are changepoints?



For data $y_1, \ldots, y_n$, a changepoint is a location $\tau$ where the statistical properties of $y_1, \ldots, y_\tau$ are different from $y_{\tau+1}, \ldots, y_n$ in some way.



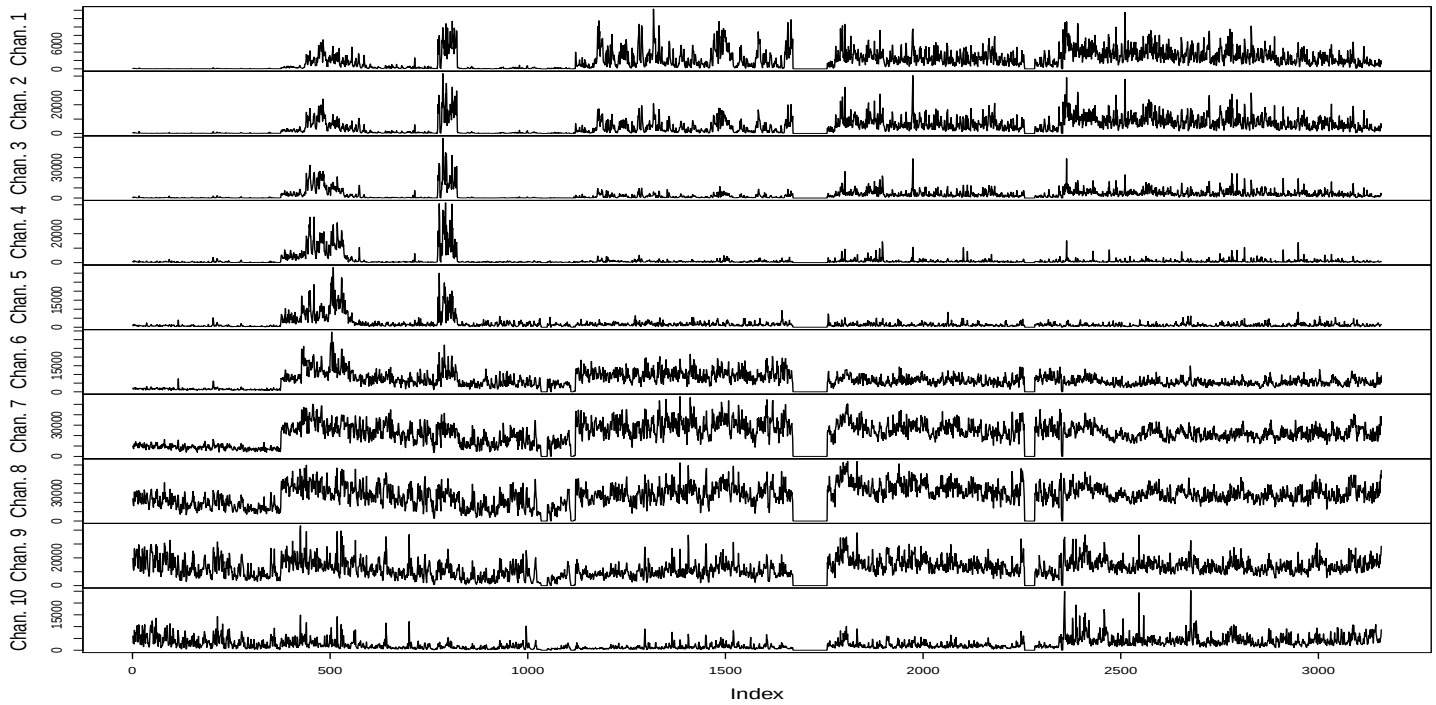




This talk is general methodology $\Longrightarrow$ no restriction on the change.

# Motivation: Distributed Acoustic Sensing

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- Vibration measurements from oil well - multiple depths over time
- Fourier-transformed data (used by engineers), sub-sampled 10x3159

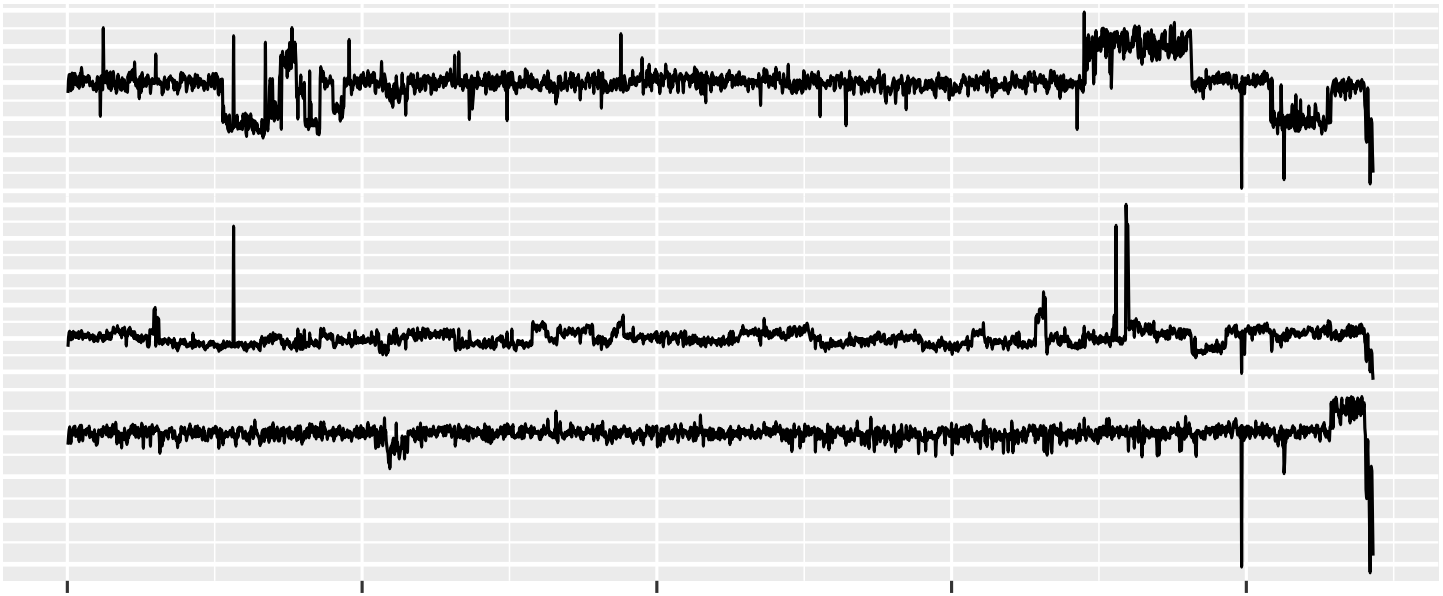



- Changes in vibration can indicate presence of impurity slugs
- Errors in measuring equipment cause striping
- True changes likely to affect only subsets

# Motivation: Genomics



- CGH measurements on DNA 43x2215
- Want to detect abnormalities common across patients with a disease



## Aim

Explicitly (and **efficiently**) identify *both* **changepoint locations** and corresponding **affected variable subsets**.

In moving to a multivariate setting a number of different scenarios can arise:

- The process in each channel could be unconnected to the rest (i.e. repeated use of univariate cpt methods might be appropriate);
- There may be some shared structure across channels. For example
  - Changes occur at the same time in all channels;
  - Changes occur in a subset of channels at the same time.
- The nature of the change could vary from one channel to another;
- . . . and doubtless many more scenarios!

# Current Approaches



**Naive Approach: Repeated Univariate Application**

Apply univariate method independently to each variable

- ✔ Explicitly identifies affected variables (but not necessarily correct)
- ✘ Omits multivariate power $\implies$ miss changes with small magnitude

# Current Approaches



**Naive Approach: Repeated Univariate Application**

Apply univariate method independently to each variable

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**Fully-Multivariate Methods**

Apply fully-multivariate method to entire (multivariate) series

- ✔ Pool information from multiple channels $\implies$ increased detection power
- ✘ Does not explicitly identify subsets (assumes all variables affected)
- ✘ Variables with no change just add noise

# Our Framework

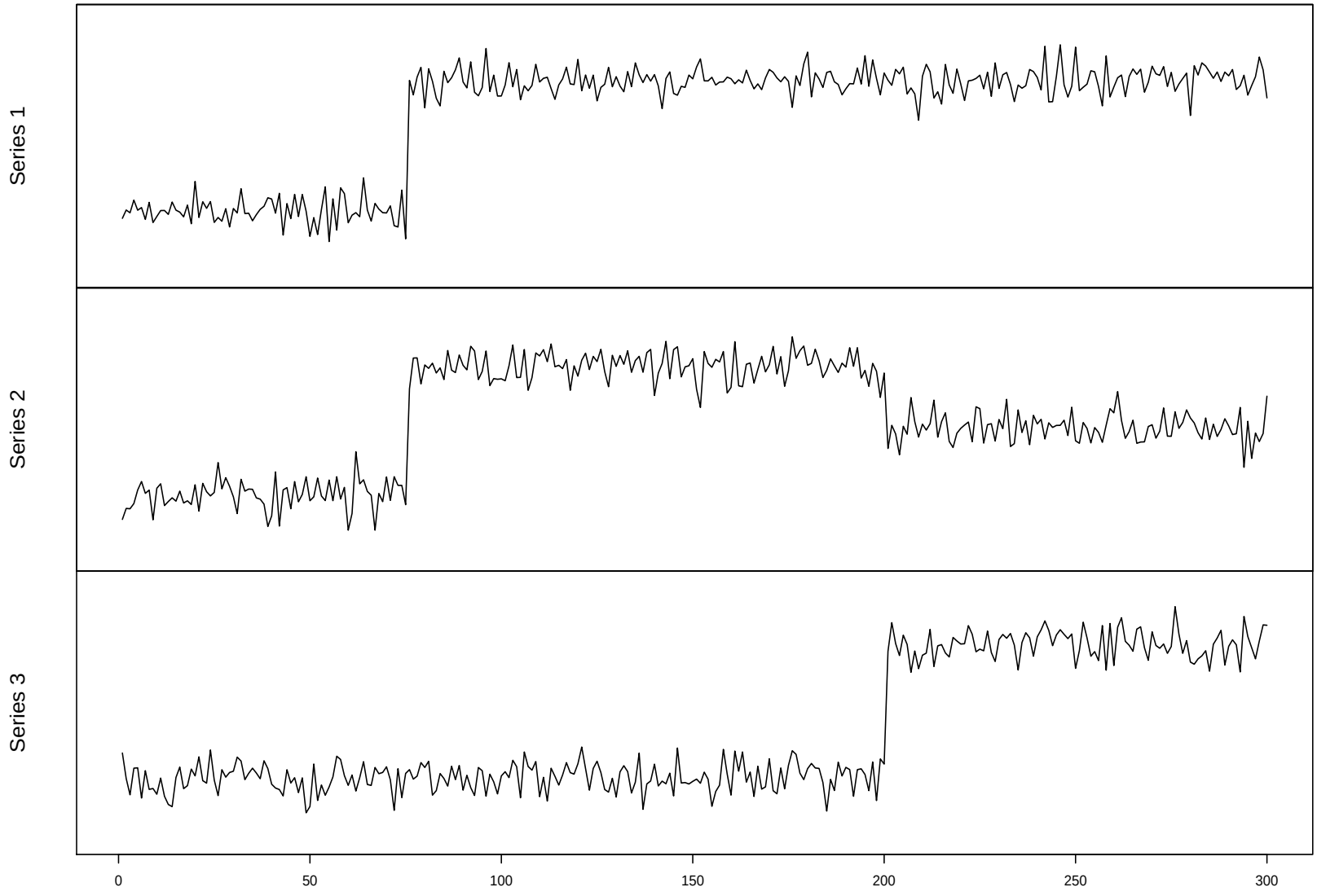


Consider a situation where the data observed within a channel are iid and there is no explicit dependence (coherence) between channels.

- Model changepoints and affected variables using single entity: **changepoint vectors**
  - Allows for segmentation of series involving subset-multivariate changes
- Formulate penalised cost for subset-multivariate segmentation using **dual penalty concept**
  - Allows separate control over univariate and multivariate detection power
- Minimise penalised cost using **Optimal Partitioning**
  - Automatically optimises number, locations and subsets of changepoints.

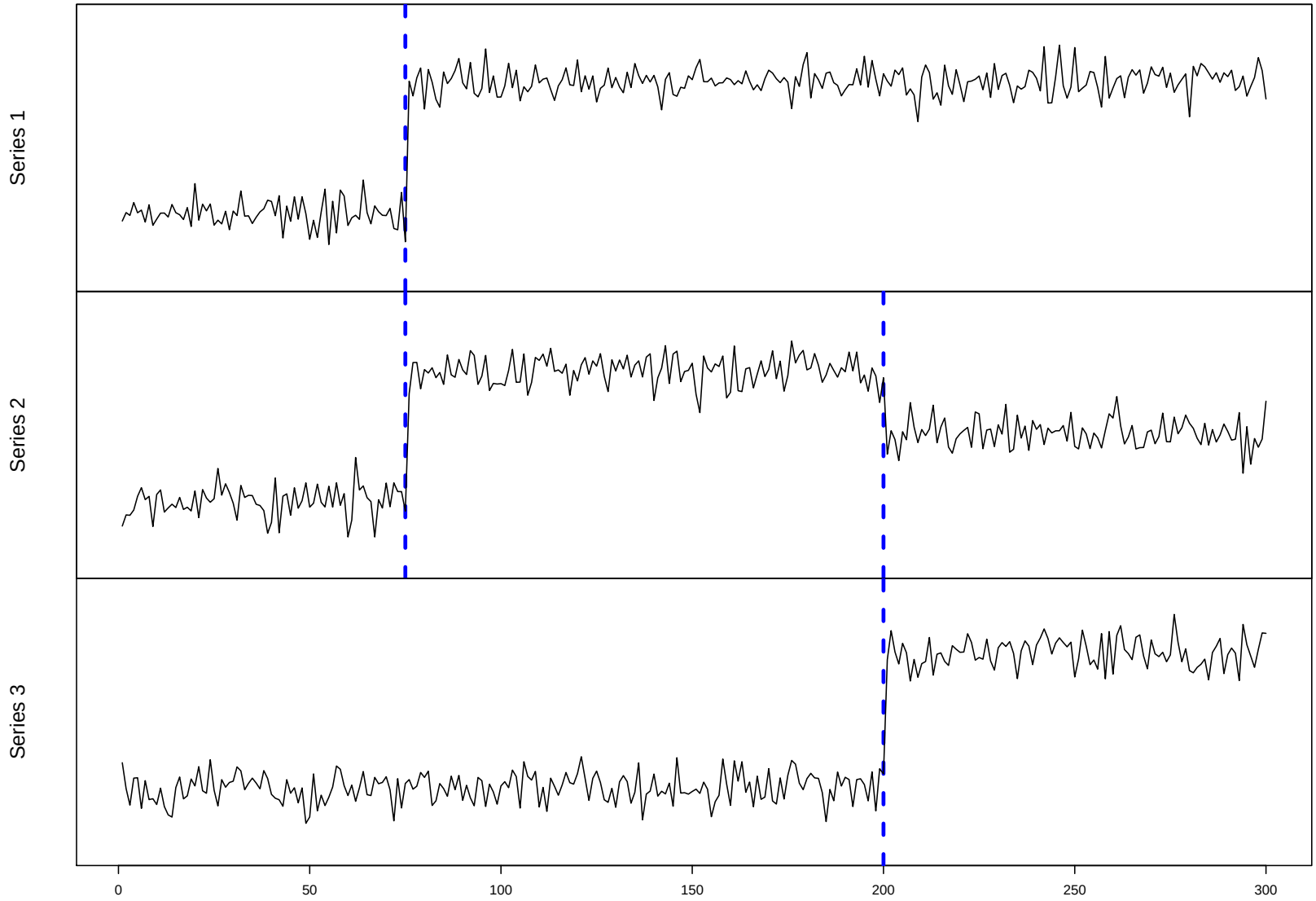
# Modelling Subset-Multivariate Cpts





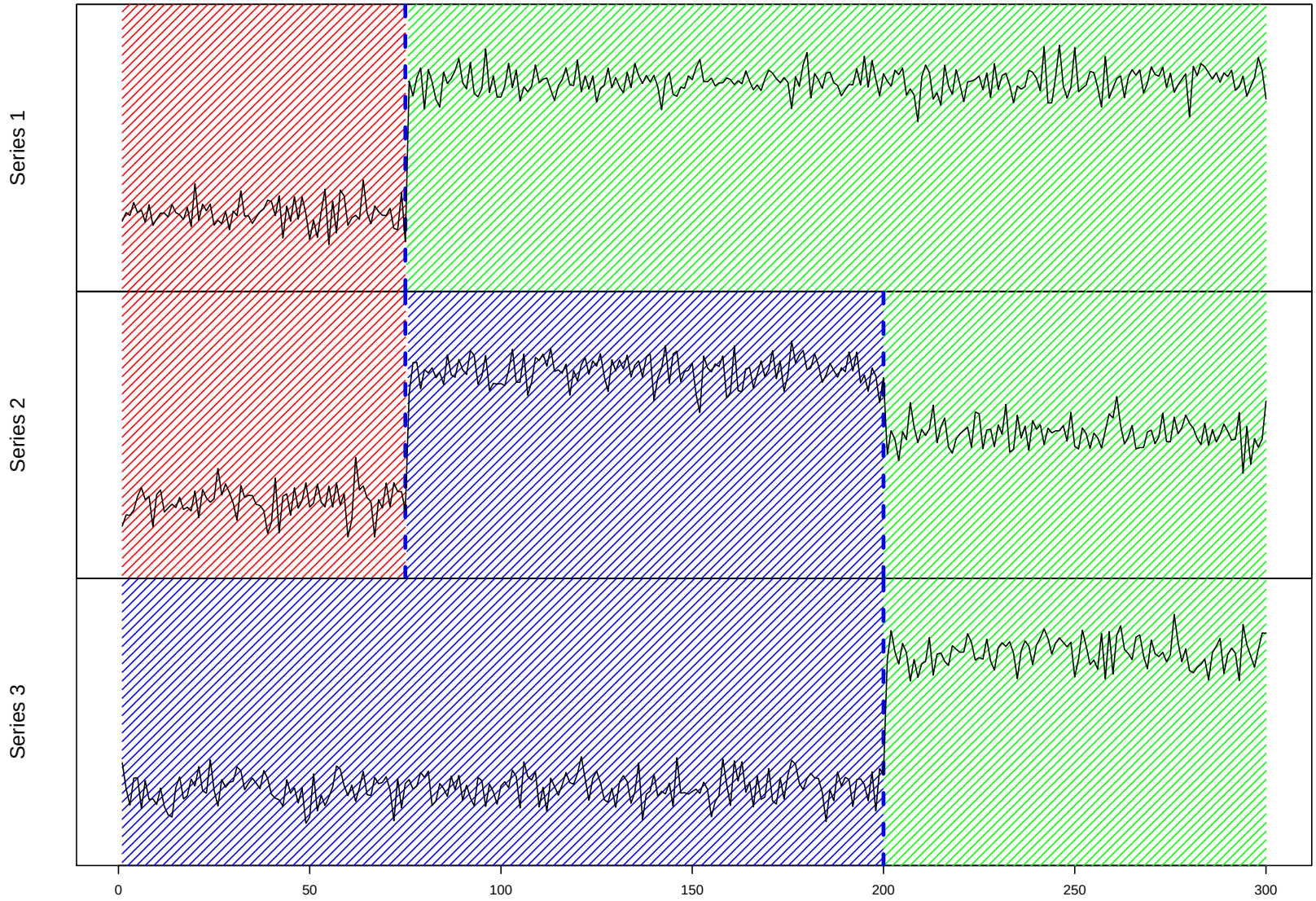
# Modelling Subset-Multivariate Cpts





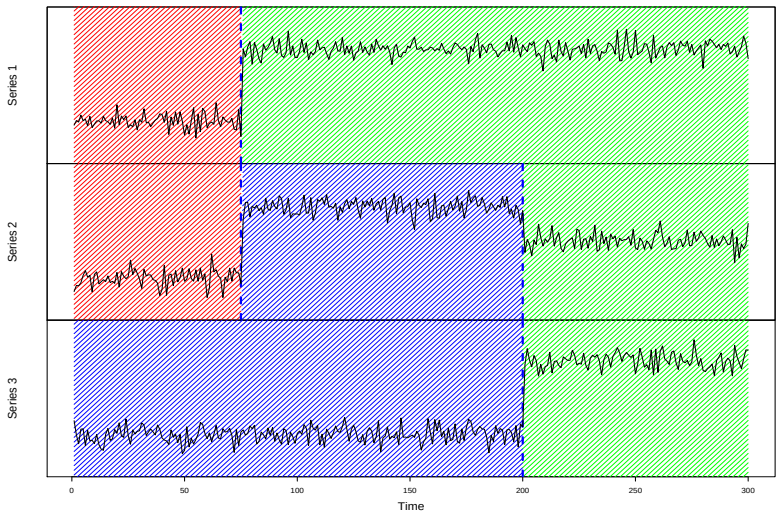
# Modelling Subset-Multivariate Cpts





# Modelling Subset-Multivariate Cpts






**Changepoints**:

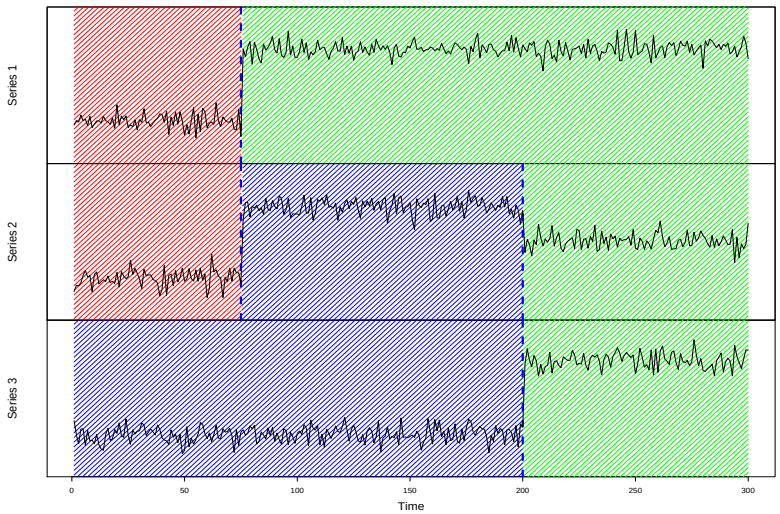
$\tau_1 = 75 \quad \tau_2 = 200 \quad \tau_3 = n = 300$

**Changepoint Vectors**:

$c_{\tau_1} = (75, 75, 0) \qquad c_{\tau_2} = (75, 200, 200)$

$c_{\tau_3} = (300, 300, 300)$

# Modelling Subset-Multivariate Cpts




**Changepoints**:

$\tau_1 = 75 \quad \tau_2 = 200 \quad \tau_3 = n = 300$

**Changepoint Vectors**:

$c_{\tau_1} = (75, 75, 0) \qquad c_{\tau_2} = (75, 200, 200)$

$c_{\tau_3} = (300, 300, 300)$

**Cost of Segmentation (assuming known changepoints)**

$$\begin{aligned}\mathcal{C}(\boldsymbol{X}) &= \left[\mathcal{C}(X^1_{(\tau_0+1):\tau_1}) + \mathcal{C}(X^2_{(\tau_0+1):\tau_1})\right] + \left[\mathcal{C}(X^2_{(\tau_1+1):\tau_2}) + \mathcal{C}(X^3_{(\tau_0+1):\tau_2})\right] \\ &\quad + \left[\mathcal{C}(X^1_{(\tau_1+1):\tau_3}) + \mathcal{C}(X^2_{(\tau_2+1):\tau_3}) + \mathcal{C}(X^3_{(\tau_2+1):\tau_3})\right] \\ &= \sum_{k=1}^{m+1}\sum_{j=1}^{p} \mathbb{I}(c^j_{\tau_k} = \tau_k)\mathcal{C}\left(X^j_{(c^j_{\tau_{k-1}}+1):c^j_{\tau_k}}\right).\end{aligned}$$

**Benefit**: Encapsulates info on changepoints *and* affected subsets

# Subset Multivariate Optimal Partitioning (SMOP)



**Aim to minimise:**

$$\sum_{k=1}^{m+1}\left\{\sum_{j=1}^{p}\left[\mathbb{I}(c^j_{\tau_k}=\tau_k)\left(\mathcal{C}_j\left(X^j_{(c^j_{\tau_{k-1}}+1):c^j_{\tau_k}}\right)+\alpha\right)\right]+\beta\right\}$$

$\mathcal{C}_j$ – Channel-Specific Cost Function

- Assigns cost to univariate data segment

$\alpha$ – Penalises increase in number of channels included in subset.

$\beta$ – Penalises increase in total number of changepoints.

# SMOP: Dynamic Program



This formulation allows us to adapt optimal partitioning (dynamic programming) for this setting $\implies$ some speed up possible.

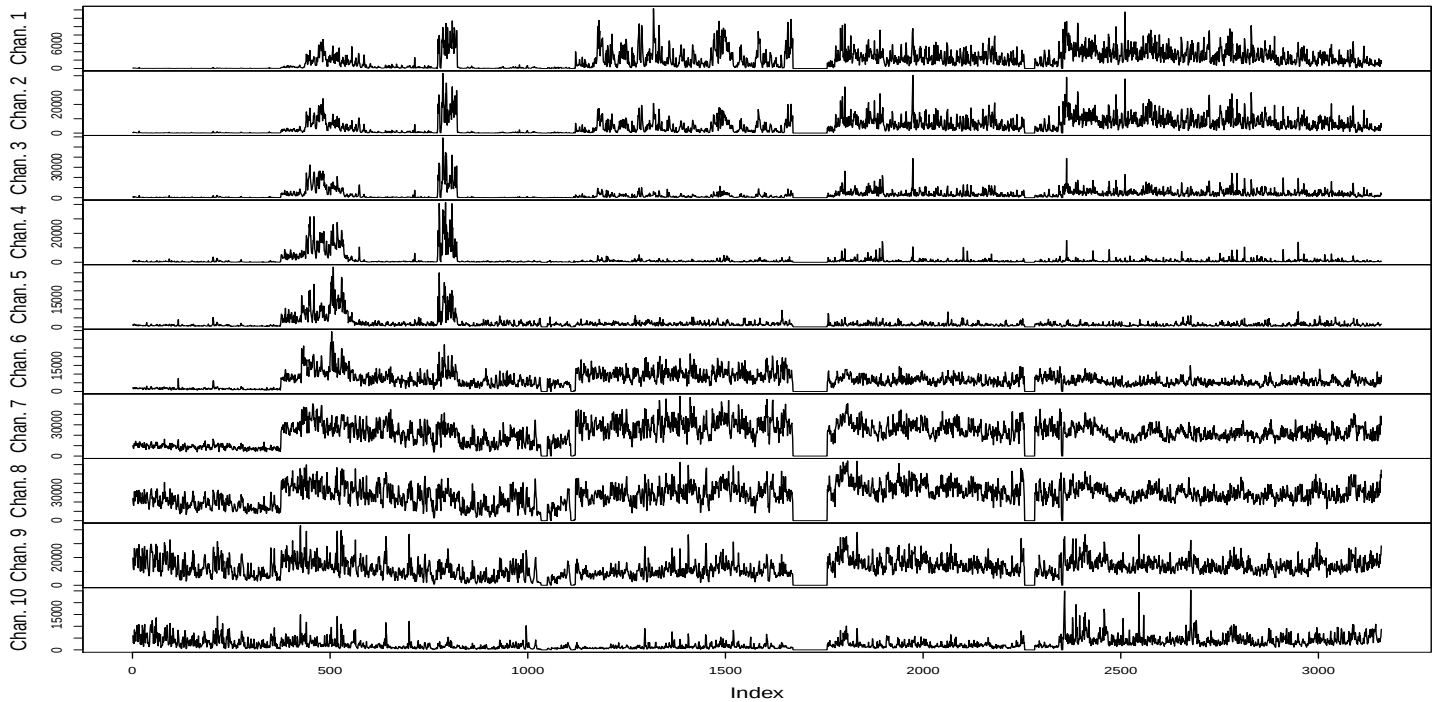
For a given changepoint vector $c_u \in C_n$ (where $u = \max(c_u)$), we have

$$F(c_u) = \min_{0 \leq t < u} \left\{ \min_{c_t \in \{\bar{C}_t \,:\, c_t^j \leq c_u^j \;\forall\; j\}} \left[ F(c_t) + \sum_{j=1}^{p} \left[ \mathbb{I}(c_t^j \neq c_u^j) \left( \mathcal{C}_j(X^j_{(c_t^j+1):c_u^j}) + \alpha \right) \right] + m(c_t, c_u)\beta \right] \right\}.$$

# Acoustic Sensing Data

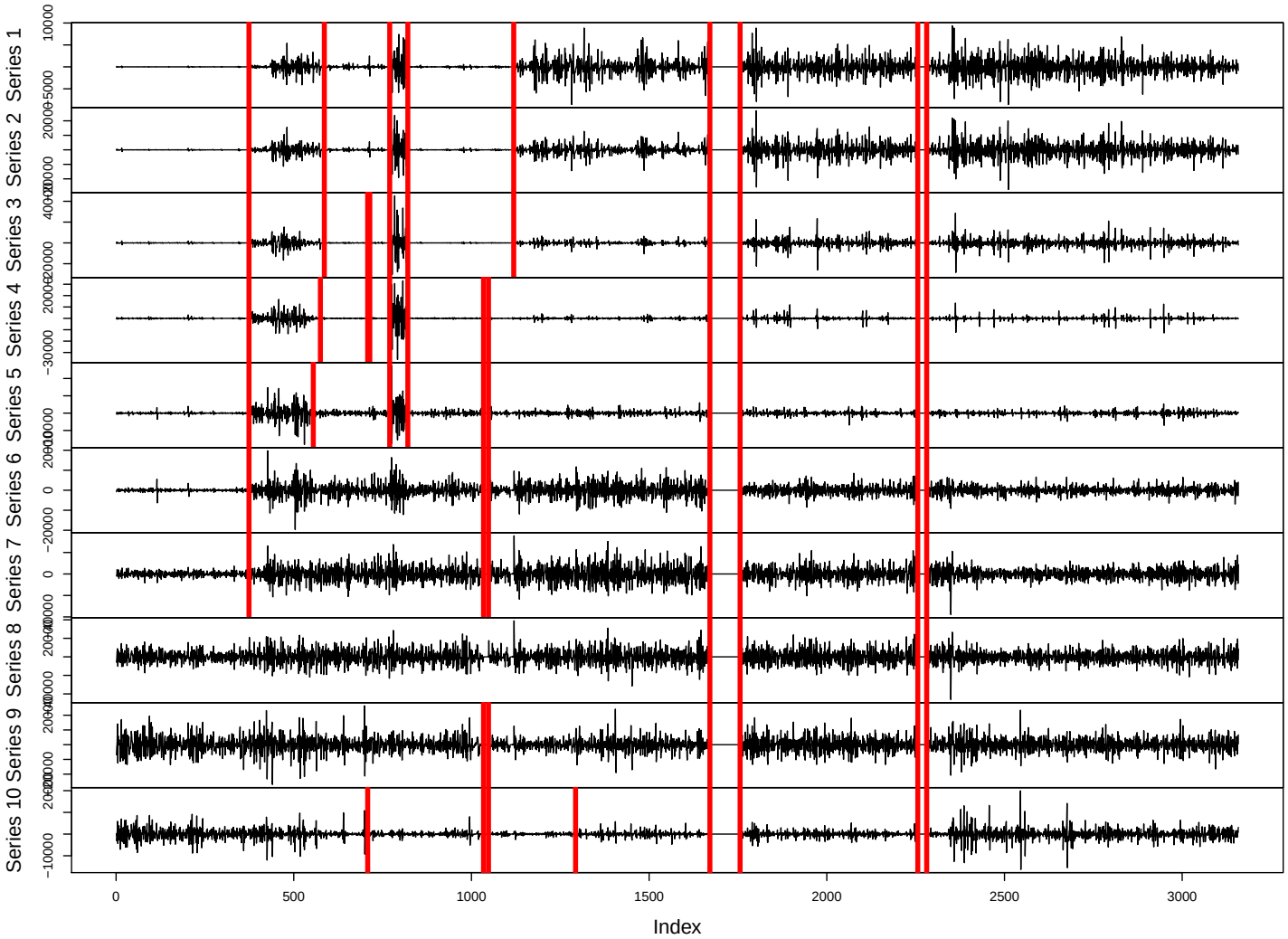


- Vibration measurements from oil well – multiple depths over time
- Fourier-transformed data (used by engineers), sub-sampled



# Acoustic Sensing Data

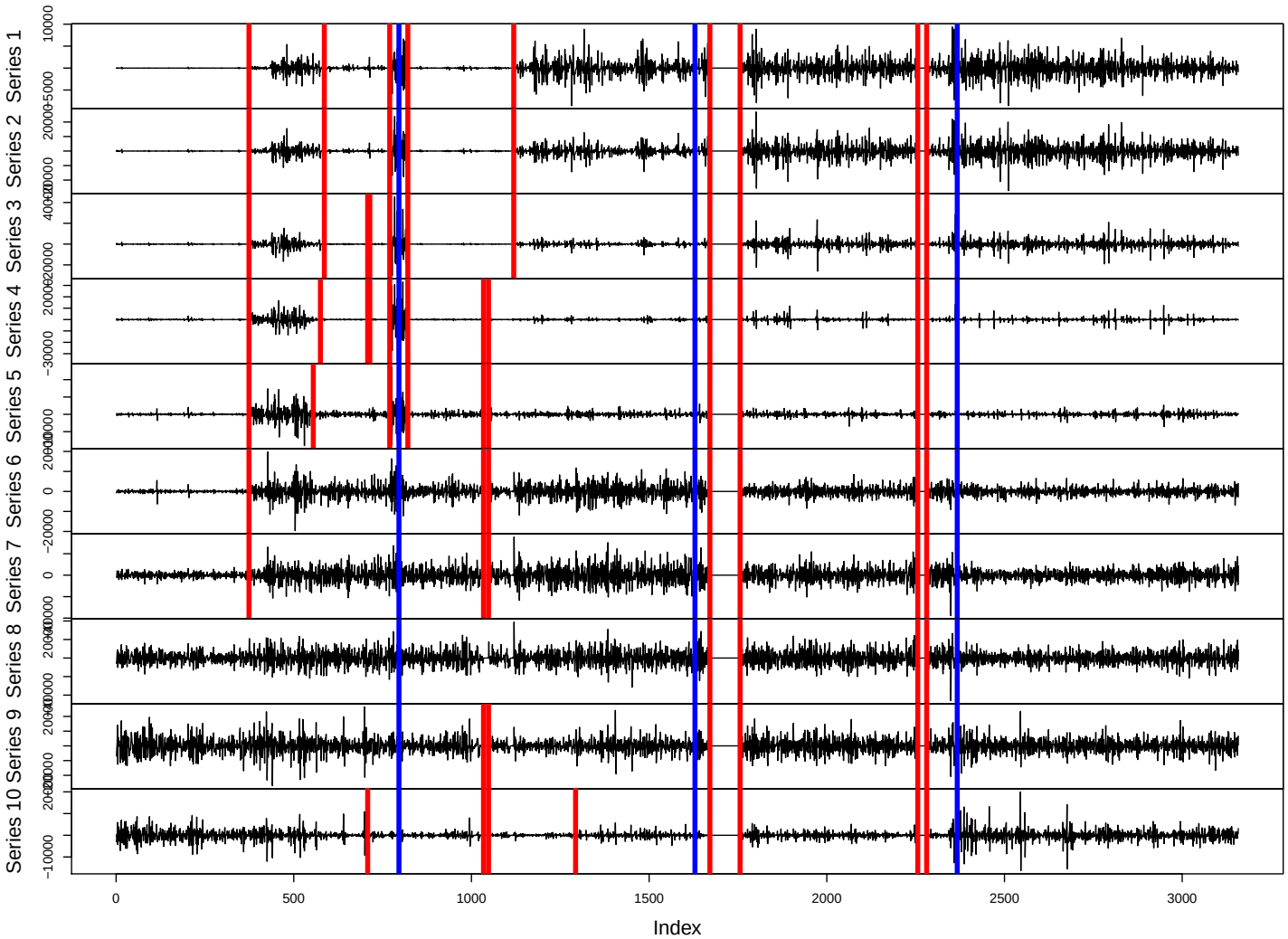





RED: SPOT

# Acoustic Sensing Data

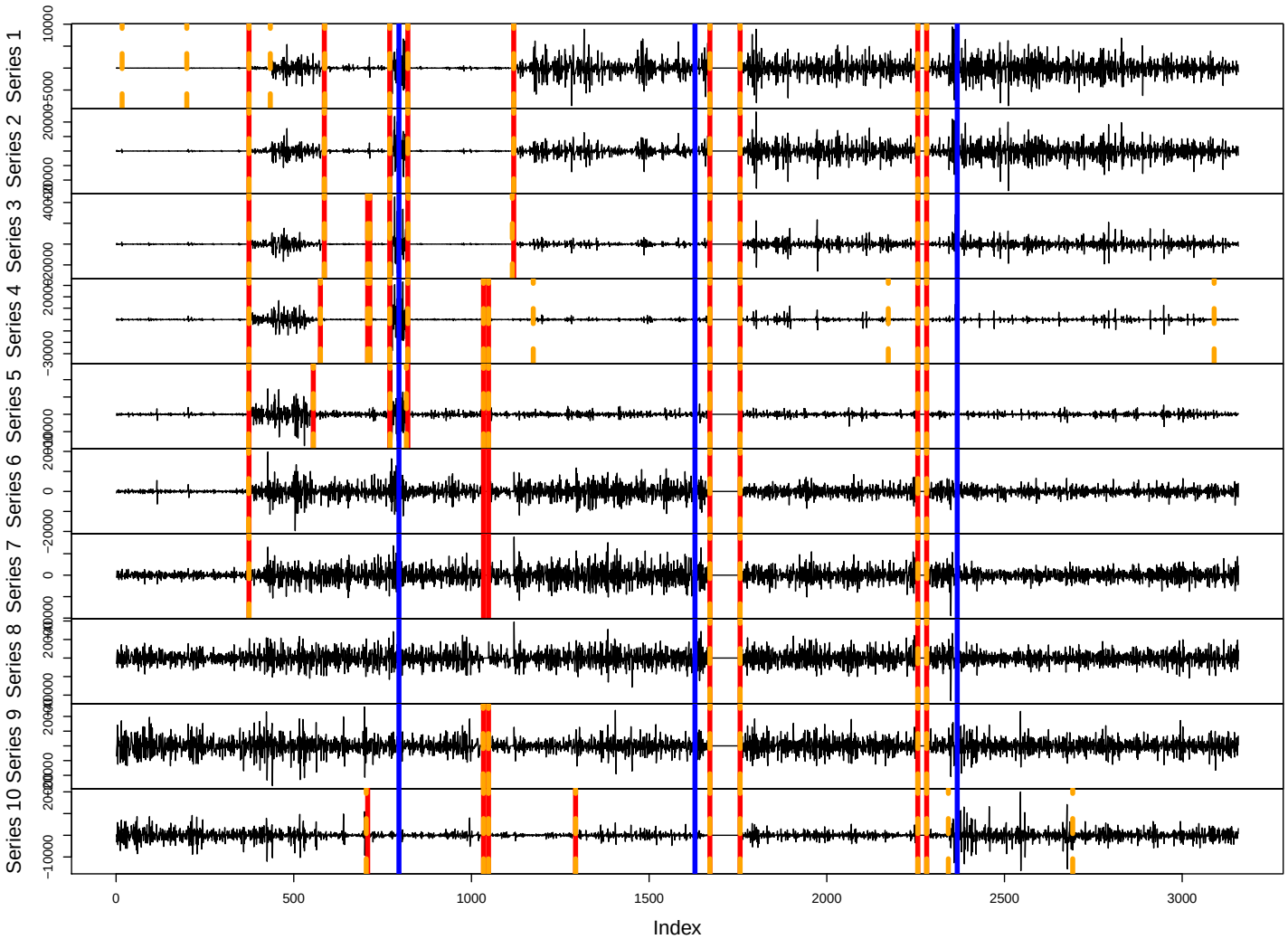





RED: SPOT BLUE: e-cp30

# Acoustic Sensing Data






RED: SPOT BLUE: e-cp30 ORANGE: univariate PELT.

# SMOP: computational considerations



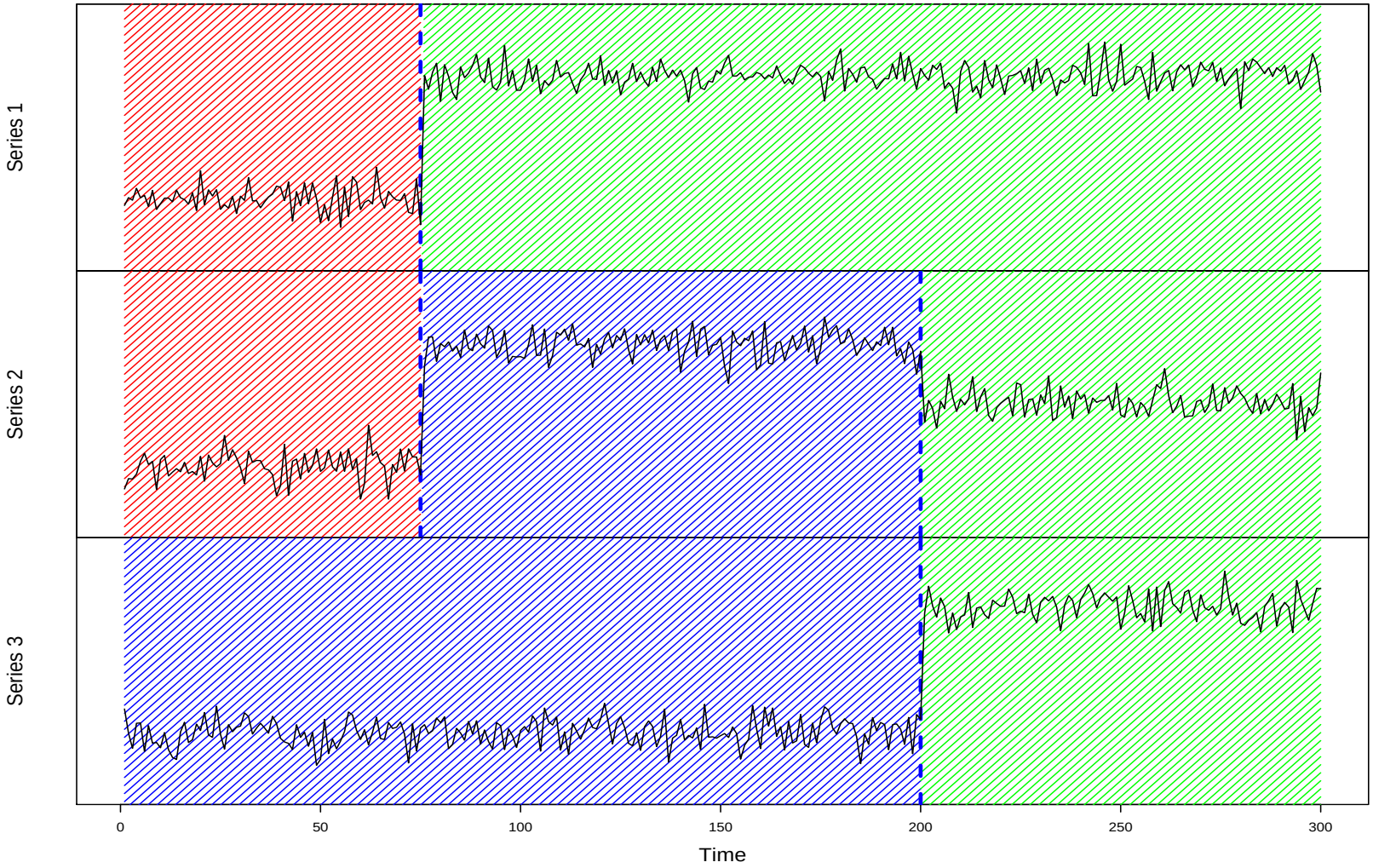
- For an exact changepoint approach, we must evaluate this penalised cost function for **all** possible changepoint locations and subsets
  $\implies$ computationally intense algorithm!
- Hence with Optimal Partitioning the **exact** approach is $\mathcal{O}(pn^{2p})$.

# SPOT

## Approximate but computationally efficient

# Approximation of SMOP





# Windowed Cost - Univariate

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$$\hat{\mathcal{C}}(p, q, w_i) = \begin{cases} \mathcal{C}(X_{p:q}|\theta(p, p + w_i)) & \text{for} \quad p - q \geq w_i \\ \mathcal{C}(X_{p:q}|\theta(p, q)) & \text{for} \quad p - q < w_i. \end{cases}$$

## Proposition

Assume that $m_0 \leq m_U$, $\theta_r^0 \neq \theta_{r+1}^0$ $(1 \leq r \leq m_0)$ and $(\tau_r - \tau_{r-1})/n$ converge to $q_r$ for $1 \leq r \leq m_0$ as $n \to \infty$. Then $Pr(\hat{m} = m_0) \to 1$ as $n \to \infty$.

# Windowed Cost - Multivariate



Natural window choice: $w_i = \tau_i - \tau_{i-1}$

$$\hat{F}(n) = \sum_{k=1}^{m+1} \left\{ \sum_{j=1}^{p} \left[ \mathbb{I}(S_k^j = 1) \left( \mathcal{C}_j \left( X^j_{(\tau_{k-1}+1):\tau_k} | \hat{\theta}^j_k \right) + \alpha \right) \right. \right.$$
$$\left. \left. + \mathbb{I}(S_k^j = 0) \left( \mathcal{C}_j \left( X^j_{(\tau_{k-1}+1):\tau_k} | \hat{\theta}^j_{k-1} \right) \right) \right] + \beta \right\}$$

$$\hat{F}(s) \approx \min_{1 \leq t < s} \left\{ \hat{F}(t) + \hat{C}(t, s | \theta_t) + \beta \right\}$$

Overall cost of this is $\mathcal{O}(pn^2)$.

# SPOT-Pruning



Works in the same way as PELT pruning (Killick et al., JASA 2012). BUT in the multivariate setting the pruning is not as efficient as in univariate.

## Pruning Rule

$$\hat{F}(t) + \hat{C}(t, s) - p\alpha \geq \hat{F}(s)$$

Then $s$ will always be a better fit than $t$ for any time $T > s$.

Theoretically we can show that for fixed $p$ the algorithm is $\mathcal{O}(n)$.

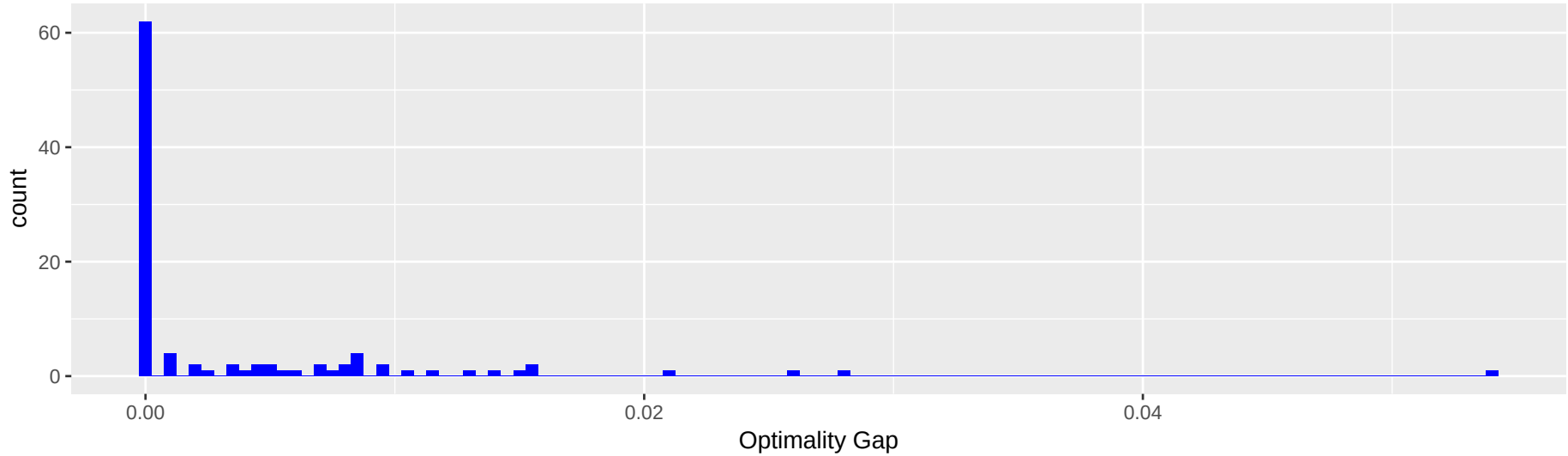
# SMOP/SPOT Comparison



Setup:

- Change in Normal mean.
- $n = 100$, $p = 3$
- 2 changes ($m = 2$)
- magnitude of change is random inside $[0.8, 1.3]$.

| Method | SPOT | SMOP |
|---|---|---|
| MSE | **0.17**(0.15) | 0.21(0.16) |
| Time (s) | **0.07**(0.01) | 15030.10(1329.5) |

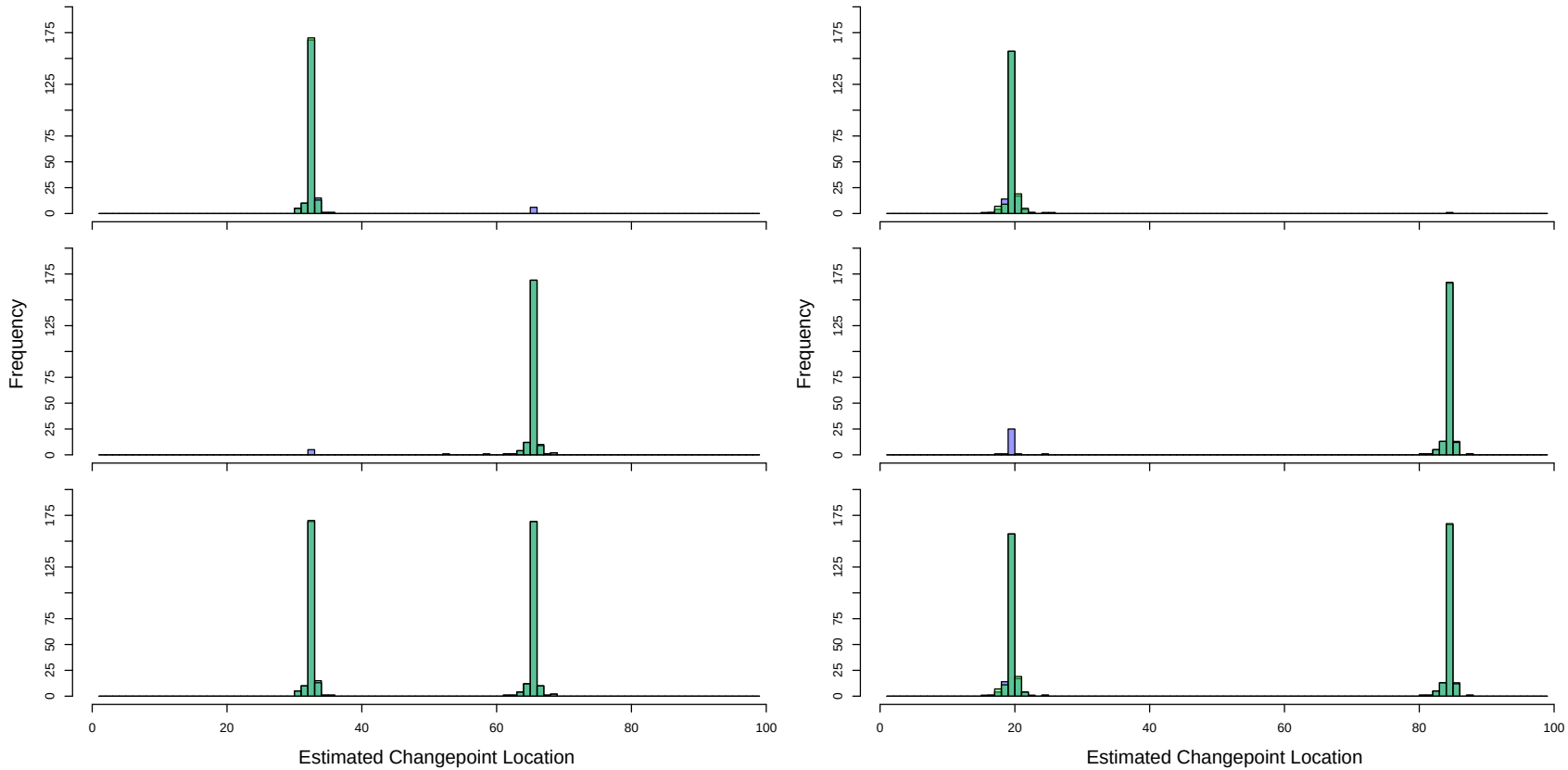


# SMOP/SPOT Comparison

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Left: Changes at 33 and 66 - almost indistinguishable

Right: Changes at 20 and 85 - demonstrating the approximation



# Simulation Examples



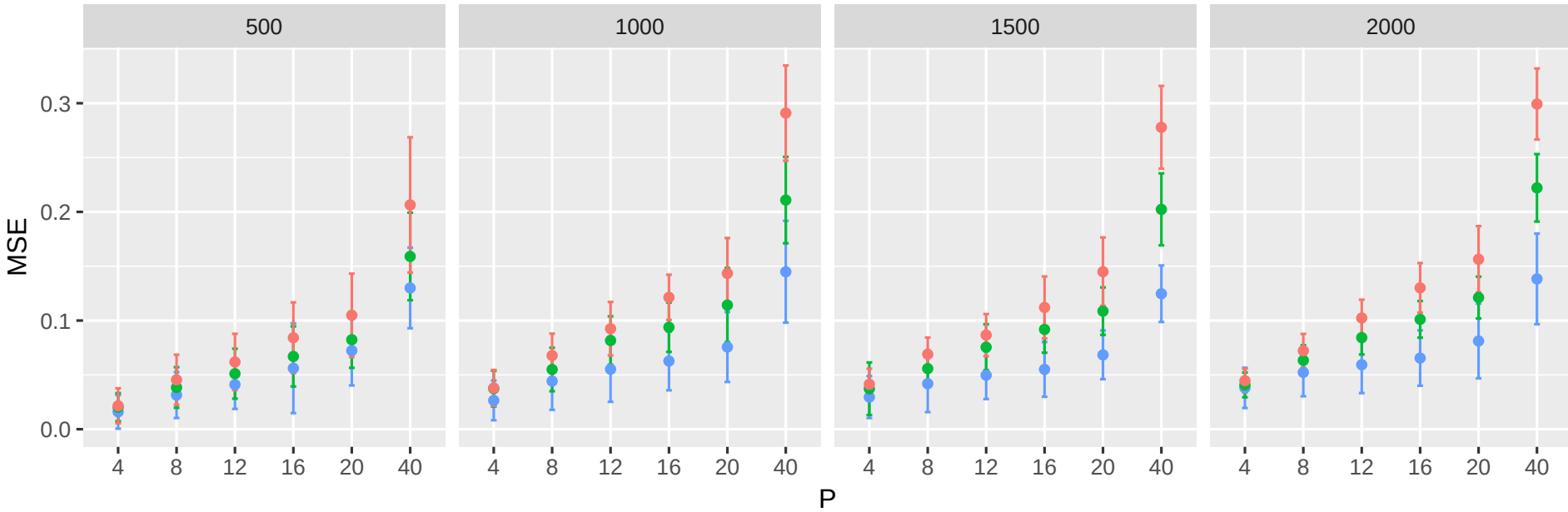
- Change in mean (so we can compare with Inspect)
- $m$ increases with $n$, $E(\tau_{i+1} - \tau_i + 1) = 200$
- Expected number of affected variates is 0.2, 0.5, or 0.8 with equal probability

Compare with the following fully multivariate approaches:

- ECP, non-parametric change in energy
- Inspect, restricted to changes in mean.

# Simulation Examples

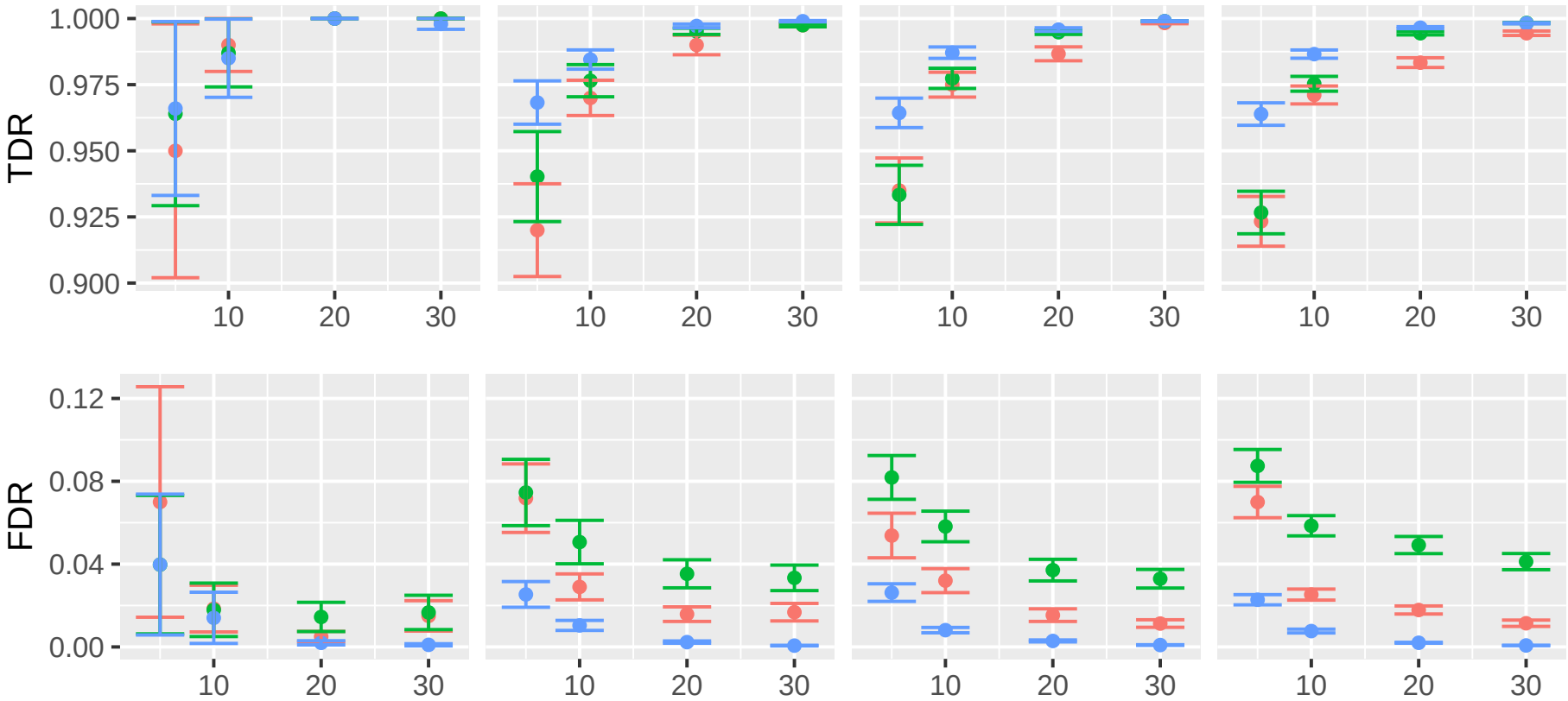





Red=ECP, Green=Inspect, Blue=SPOT

# Simulation Examples

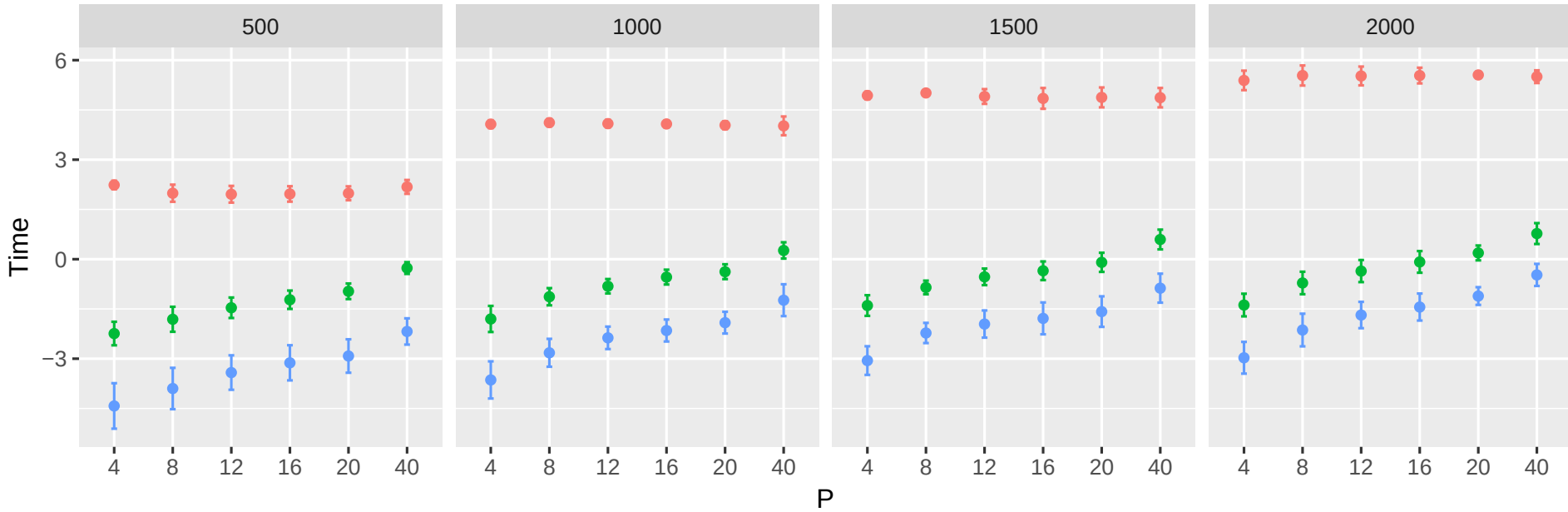





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# Simulation Examples

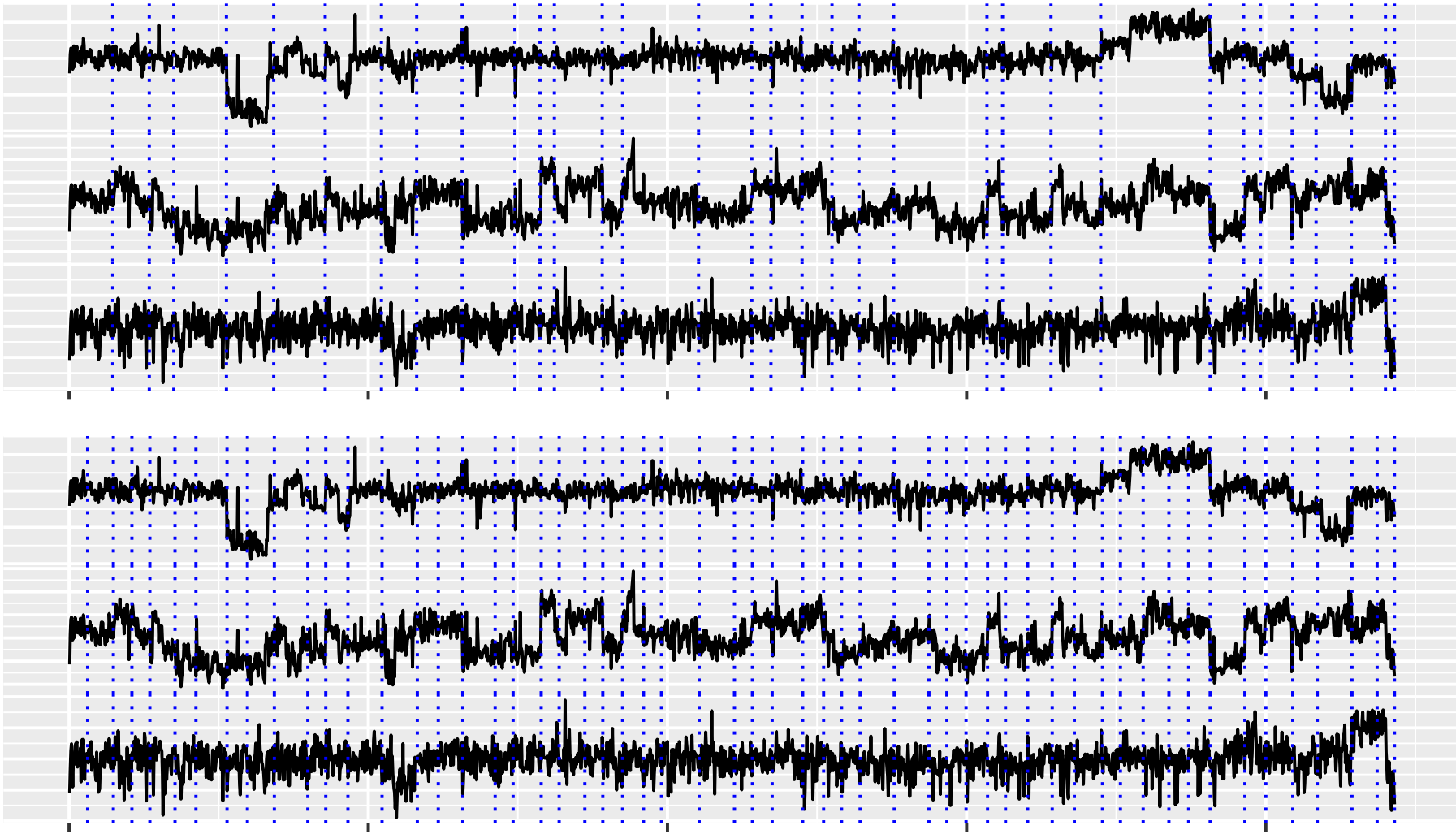





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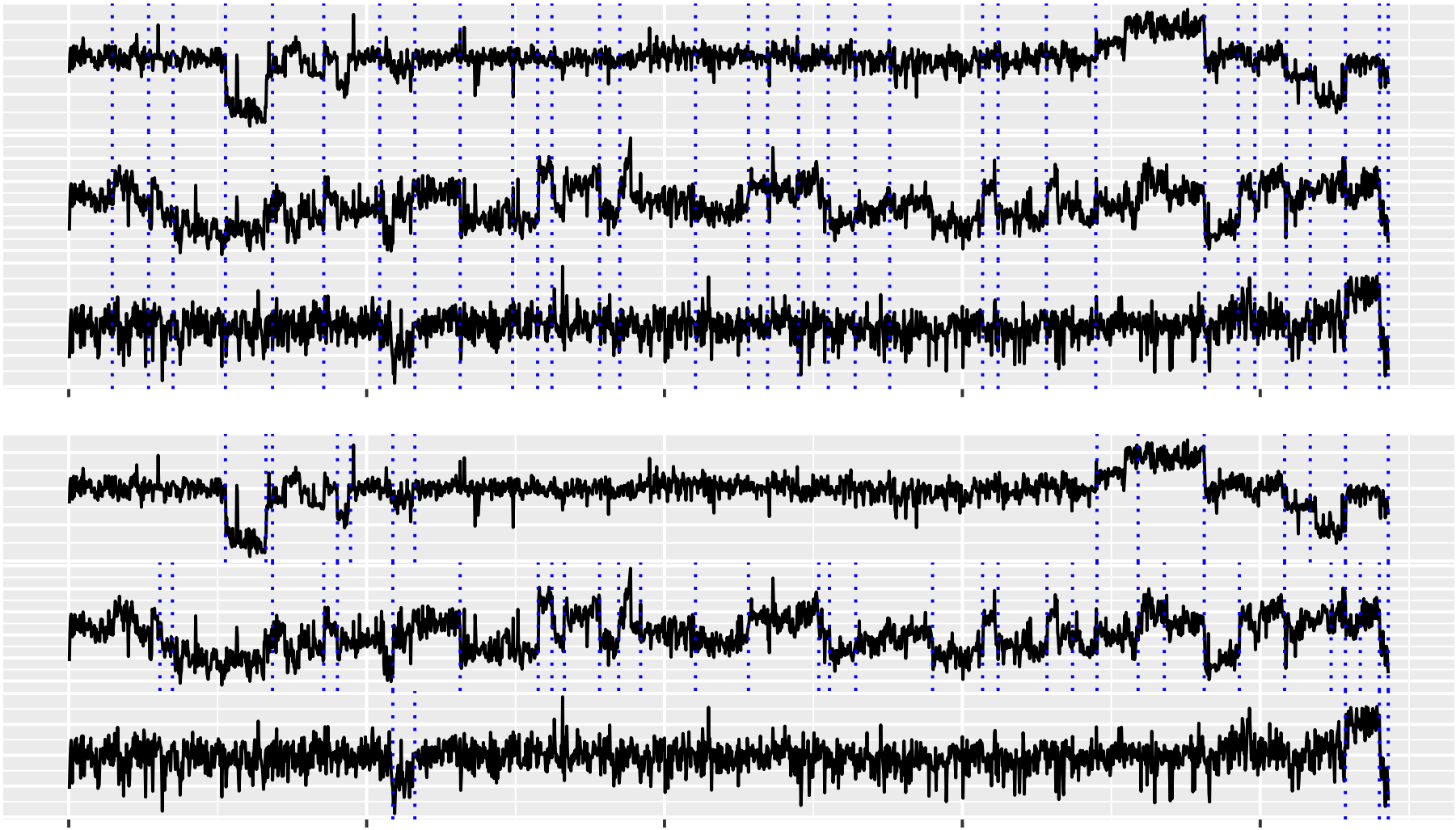
# CGH results: Inspect & ECP





# CGH results: Inspect & SPOT





# CGH results



- Inspect paper: 30 most significant changepoints
- ECP default: 54 changepoints
- SPOT default: 69 changepoints

- But for SPOT only 10 affect more than 25 series out of 43

# Syrian Civil War



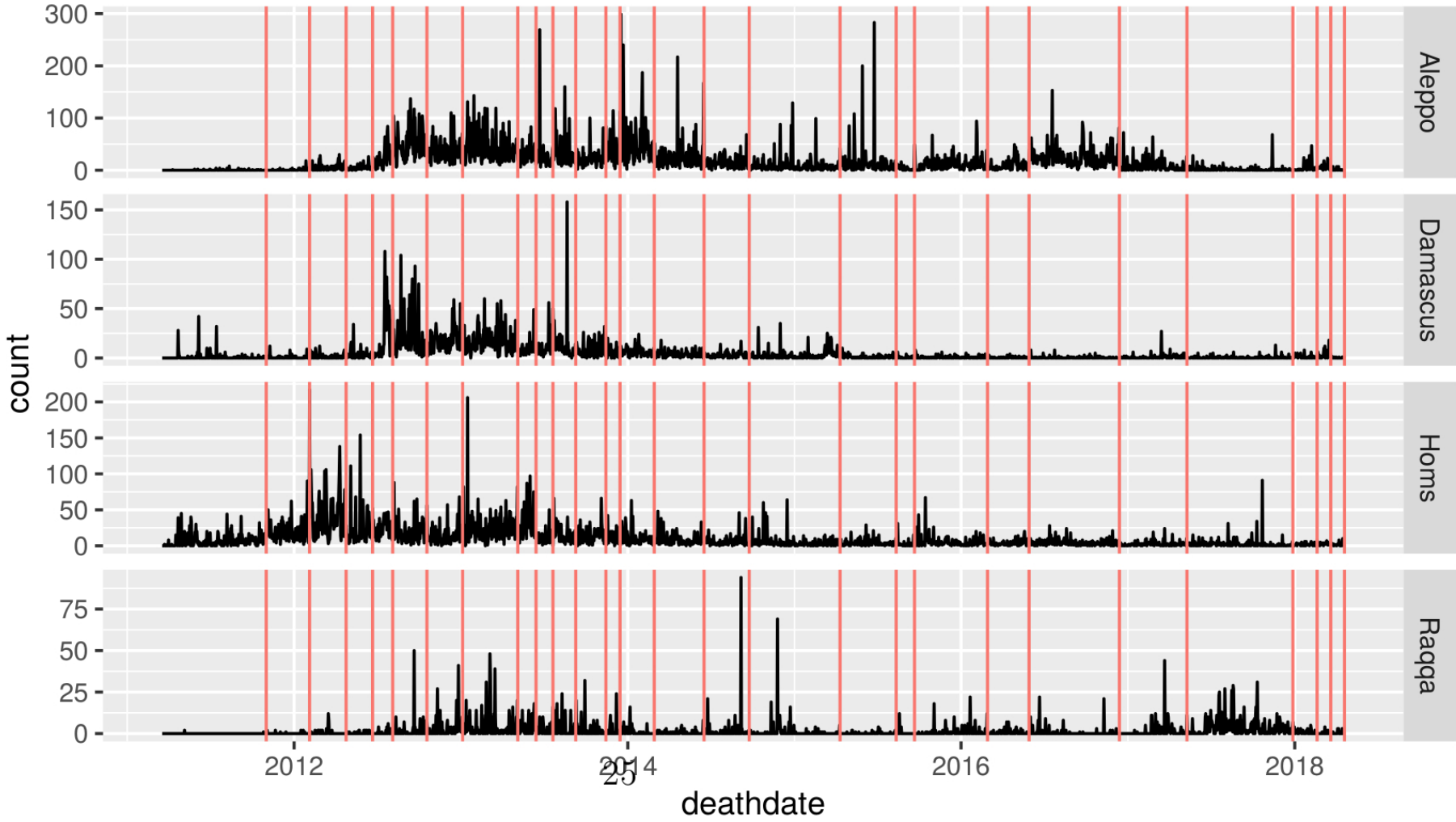
- Violations Documentation Centre
- Records confirmed deaths

- Name of victim
- Date and region where they died

- Discrete, bounded support

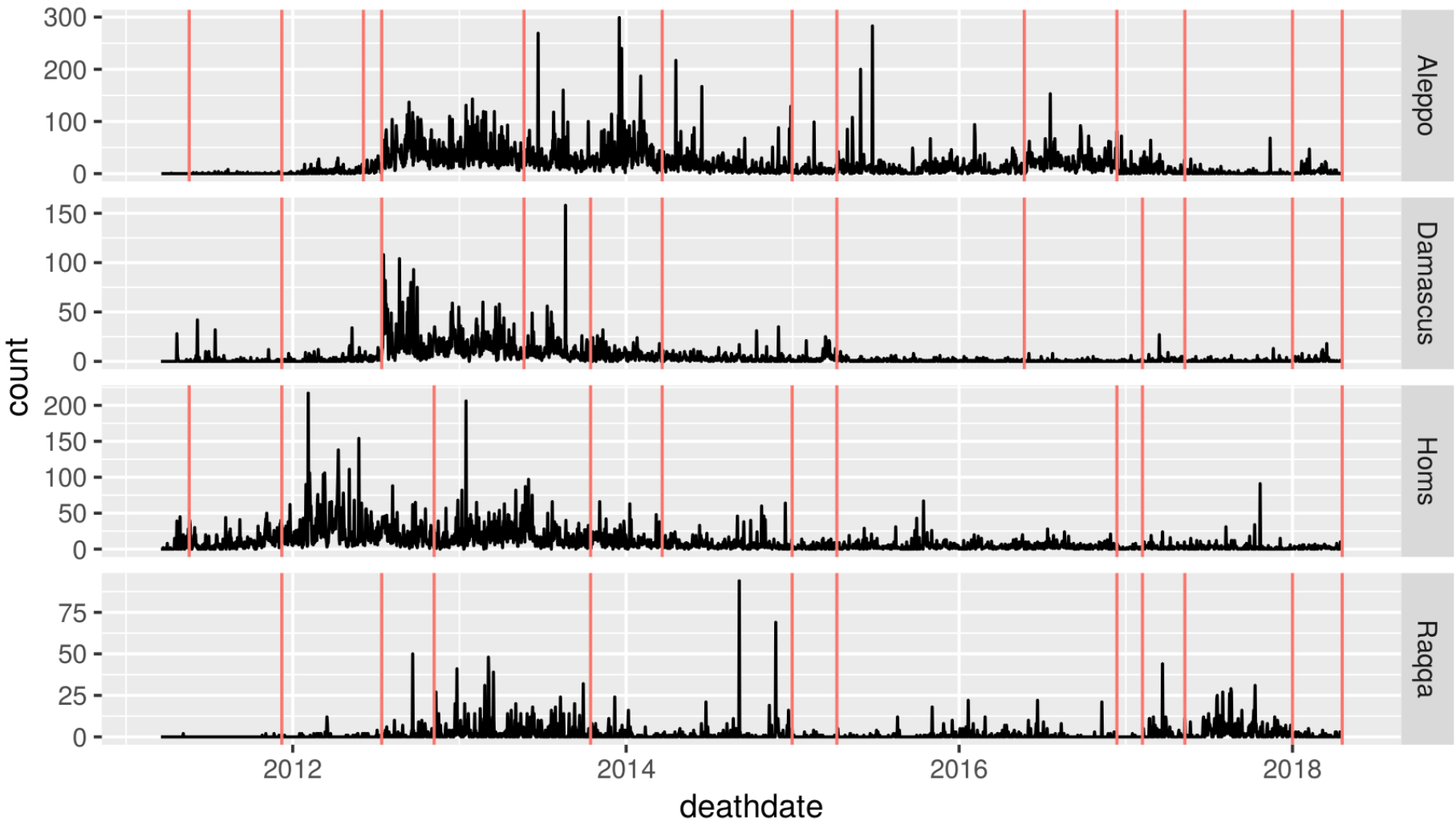
# Syrian results: ECP





# Syrian results: SPOT





# Summary



- Identifying which subsets of multivariate series undergo a change is important in many practical applications
- SMOP: A promising subset multivariate changepoint detection framework
- SPOT: using windows to approximation SMOP permits good segmentation accuracy in comparatively much lower computational time
- SPOT and SMOP available in `changepoint.mv` R package to be updated on CRAN shortly

- Invite others to use the SMOP framework and devise either approximations or computationally efficient methods for exact minimizations.